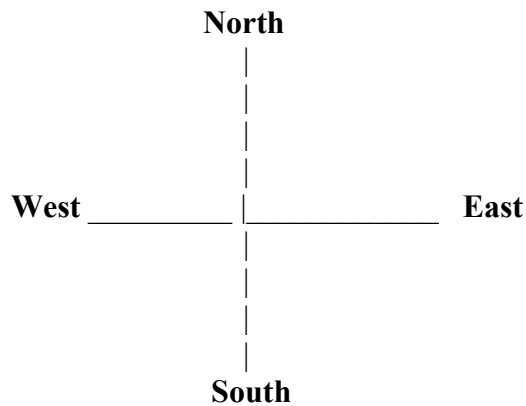


Locating Places on the Earth

Maps

- A map is a drawing or representation of an area seen from above.
- Important Components of Maps are **Distance, Direction and Symbols** .
- Maps use a **scale** to represent real distances.
Like 1cm =300m.



Types of Maps

1. **Physical Maps** – show mountains, rivers, oceans, etc.
2. **Political Maps** – show countries, states, cities, boundaries.
3. **Thematic Maps** – show specific information like rainfall or population.

Globe:- A globe is a spherical model of the Earth. It represents Earth more accurately than flat maps.

Coordinates(Latitudes and Longitudes)

Latitude

Imaginary lines running east-west parallel to the Equator.

Measure distance from the Equator.

Important Latitudes

Equator = 0°

North Pole = 90°N

South Pole = 90°S

Climate and Latitude

Near Equator → Hot (Torrid zone)

Middle regions → Moderate (Temperate zone)

Near poles → Cold (Frigid zone)

Longitude

Imaginary lines running from North Pole to South Pole.

Measure distance from the Prime Meridian.

Prime Meridian

Located at Greenwich, London.

Marked as 0° longitude.

Longitude Range

0° to 180° East (E)

0° to 180° West (W)

Hemispheres

Equator divides Earth into:

Northern Hemisphere

Southern Hemisphere

Prime Meridian divides Earth into:

Eastern Hemisphere

Western Hemisphere

Time and Time Zones

Earth's Rotation

Earth rotates from west to east.

Completes one rotation in 24 hours.

Relation Between Longitude and Time

360° = 24 hours

15° longitude = 1 hour

Local Time

Time based on the Sun's position at a particular place.

Standard Time

Common time used across a country.

Indian Standard Time (IST)

IST is based on 82.5°E longitude.

IST = GMT + 5 hours 30 minutes

Time Zones

Earth is divided into different time zones.

Large countries like: Russia ,USA and Canada have multiple time zones.

International Date Line (IDL)

Located near 180° longitude.

Crossing it changes the date:

Eastward → subtract one day

Westward → add one day